5-MINUTE REVISION

Data Transformation

From the chapter's own revision points, formulae and questions.

Selected from Chapter 24 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- Data transformation → used when the assumptions required for valid statistical analysis are not satisfied.
- Main purpose → improve validity, precision, and accuracy of statistical analysis.
- Transformation should not be chosen merely to obtain a desired conclusion.
- ANOVA and mean separation → performed on the transformed data.
- Final means → retransformed to the original scale for interpretation.

Quick revision · 2

- Normality → assess model errors using residual diagnostics after transformation.
- Transformations covered → logarithmic, square root, angular, and the Box-Cox family.
- Logarithmic transformation → suitable when standard deviation is proportional to the mean.
- Log transformation → also useful for multiplicative rather than additive effects.
- Logarithms require positive inputs; shifts need justification, and an alternative model may be preferable.

Quick revision · 3

- Common transformation for zero values → $\log(Y+1)$.
- Positive observations below one can be logged directly; their logarithms are negative.
- Square root transformation → appropriate for Poisson-type count data.
- Examples → insect counts, number of defectives, and counts of seeds affected by disease.
- Poisson distribution → $\sigma^2 = \mu$.

Quick revision · 4

- Empirical square root case → $\sigma^2 = c^2\mu$.
- Zero counts → $\sqrt{Y}$ is defined; the worked example uses $\sqrt{Y+1}$ consistently.
- Retransformation of square root mean → square the value and subtract 1.
- Retransformed square root mean → no greater than the original arithmetic mean; the difference may be substantial.
- Back-transformed means are not generally arithmetic means; any bias correction must match the fitted model.

Quick revision · 5

- Angular transformation → appropriate for percentage data derived from binomial counts.
- Binomial variance → $\sigma^2 = p(1-p)/n$.
- The 30%–70% guideline is approximate; use the binomial model and residual diagnostics to assess the analysis.
- A single proportion outside 30%–70% does not automatically require transformation. If used, apply one transformation consistently.
- Angular transformation → $sin^{-1}\sqrt{p}$.

Quick revision · 6

- For 0% → replace with $1/(4n)$.
- For 100% → replace with $1 - 1/(4n)$.
- Retransformation of angular mean → $100 \times sin^2 X$, using the same angular unit as in the forward transformation.
- Percentage of protein, carbohydrate, etc. → need not necessarily be transformed because such percentages are not necessarily binomial proportions.
- Distribution and transformation:
  - Poisson → $\sqrt{x}$.
  - Empirical $\sigma^2 = c^2\mu$ → $\sqrt{x}$.
  - Binomial → $sin^{-1}\sqrt{p}$.
  - Empirical $\sigma^2 = c^2\mu^2$ → $\log(x)$.

Quick revision · 7

- Logarithmic example → weed counts were transformed using $X = \log Y$.
- Square root example → larval counts containing zero values were transformed using $X = \sqrt{Y+1}$.
- Angular example → insect mortality percentages were transformed using $X = sin^{-1}\sqrt{p}$.
- In all three examples → ANOVA was performed on transformed values.
- Treatment means → compared using the appropriate critical difference after significant treatment F test.

Quick revision · 8

- Final interpretation → based on retransformed means expressed in the original units.

Important formulae · 1

Logarithmic transformation:

$$X = \log Y$$

Logarithmic transformation with a constant:

$$X = \log(Y + k), \quad k > 0$$

Square root transformation:

$$X = \sqrt{Y}$$

Important formulae · 2

Square root transformation when zero values occur:

$$X = \sqrt{Y+1}$$

Alternative transformation for very small counts:

$$X = \sqrt{Y} + \sqrt{Y+1}$$

Angular transformation:

$$X = sin^{-1}\sqrt{p}$$

where

$$p = \frac{Y}{100}$$

Important formulae · 3

Replacement for 0%:

$$p = \frac{1}{4n}$$

Replacement for 100%:

$$p = 1 - \frac{1}{4n}$$

Retransformation after logarithmic transformation:

$$Y = antilog X$$

Important formulae · 4

Retransformation after $X = \sqrt{Y+1}$:

$$Y = X^2 - 1$$

Retransformation after angular transformation:

$$Y = 100 \times sin^2 X$$

Binomial variance:

$$\sigma^2 = \frac{p(1-p)}{n}$$

Important formulae · 5

Poisson variance:

$$\sigma^2 = \mu$$

Empirical variance relationship for square root transformation:

$$\sigma^2 = c^2\mu$$

Empirical variance relationship for logarithmic transformation:

$$\sigma^2 = c^2\mu^2$$

Important formulae · 6

Coefficient of variation:

$$CV = \frac{SD}{Mean} \times 100$$

Critical difference for transformed means in an RBD:

$$CD = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{r}}$$

Self-check from the chapter

1. A change of measurement scale used before analysis is called a data _____. **[Transformation]**
2. Unequal error variances are described as _____. **[Heteroscedasticity]**
3. A transformation commonly used for positive data with multiplicative effects is the _____ transformation. **[Logarithmic]**
4. A traditional variance-stabilising transformation for Poisson counts is the _____ root transformation. **[Square]**